

Name: _____

Class: _____

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS HSC ASSESSMENT TASK 3

JUNE 2006

Time Allowed: 70 minutes

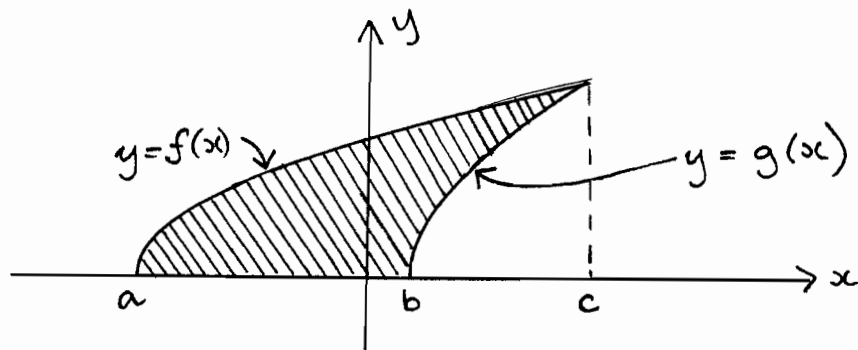
Instructions

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination, this examination paper must be attached to the front of your answers.
- Attempt all questions.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- A table of standard integrals is supplied.

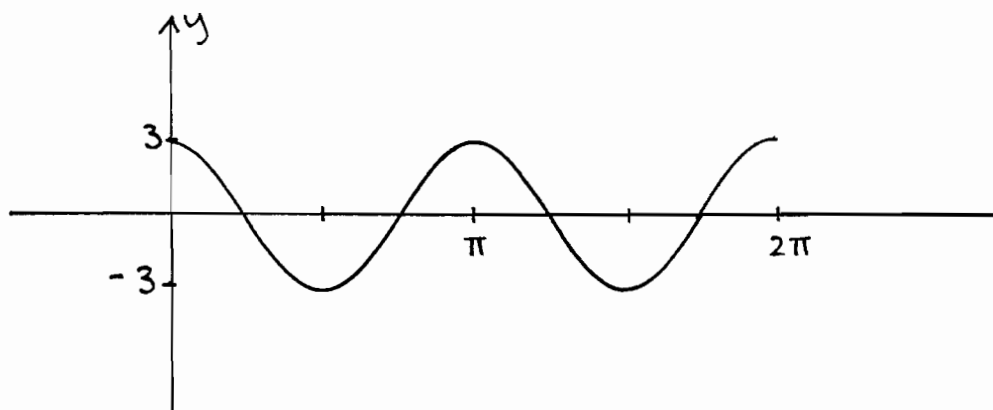
Q1	Q2	Q3	Q4	Q5	Total
11	12	12	12	10	57

Question 1 (11 Marks)

- a) Find the exact value of $\sin \frac{2\pi}{3}$ 1
- b) Find $\cos 1.5^\circ$ correct to 3 decimal places. 1
- c) Express 2.25π radians in degrees. 1
- d) Find $\lim_{x \rightarrow 0} \frac{\sin x}{2x}$ 1
- e) Express the shaded area below as either the sum or difference of two integrals (correct notation must be used) 2



- f) The curve below has been drawn from $x = 0$ to $x = 2\pi$. The curve has equation in the form $y = a \cos bx$. Find a and b . 2

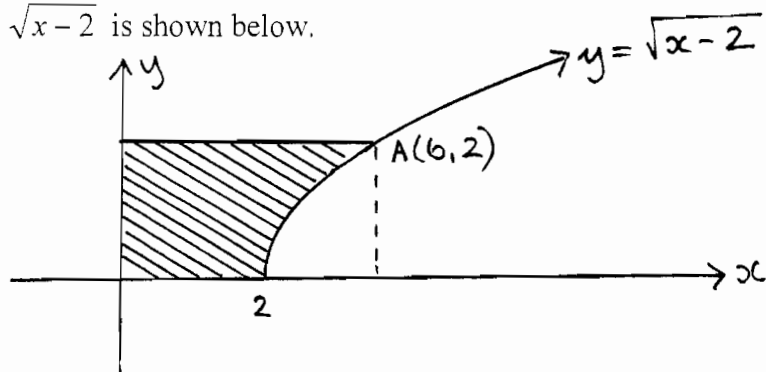


- g) Draw a neat sketch of $y = f(x)$ in the domain $a \leq x \leq b$ given that $f'(x) > 0$ and $f''(x) > 0$ in the domain and $f(a) = 0$ 3

Question 2 (start a new page) (12 marks)

- a) The curve $y = \sqrt{x-2}$ is shown below.

3



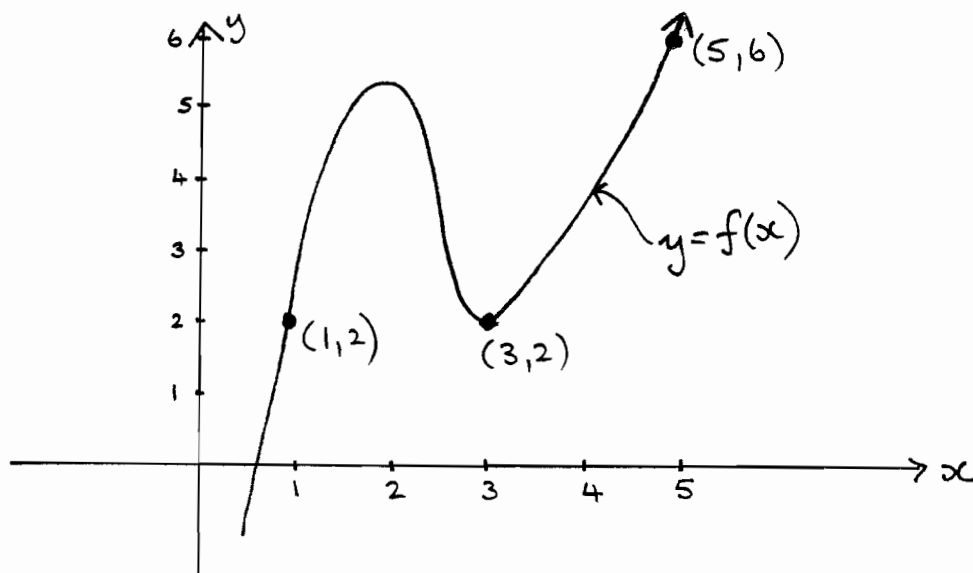
A(6,2) lies on the curve.

Find the shaded area.

- b) i) Find the approximate area enclosed by the curve $y = f(x)$, the x axis and the lines $x = 1$ and $x = 5$, by using 3 function values and the Trapezoidal Rule.

The curve $y = f(x)$ is shown below.

2



- ii) Is your answer in part i) an under or over estimate of the exact area.
Explain your answer.

1

- c) The curve $y = \sqrt{\cos \pi x}$ from $x = 0$ to $x = \frac{1}{2}$ is rotated around the x axis. What is the volume of the solid of revolution generated?

3

- d) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x \, dx$

3

Question 3 (start a new page) (12 marks)

- a) Solve $2 \cos^2 x + 3 \cos x - 2 = 0$ for x , if $0 \leq x \leq 2\pi$

3

- b) i) Find $\frac{d}{dx}(\sin^2 x)$

2

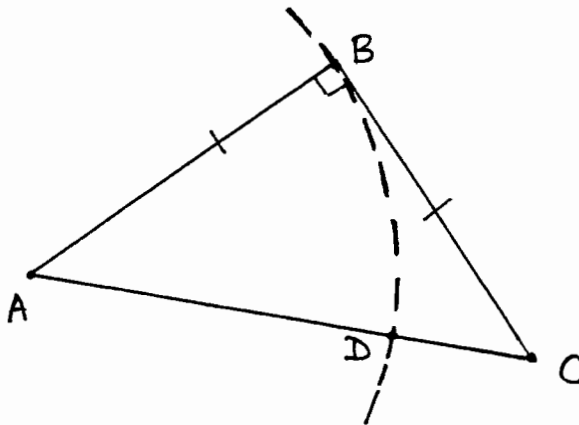
- ii) Find $\frac{d}{dx}(\sin x \cdot \cos 2x)$

2

- iii) Find $\int \sin(2x+1) \, dx$

2

- c) ABC is an isosceles right angled triangle. $AB=BC=4\text{cm}$. An arc, centre A and radius 4cm is drawn to cut the side AC at D.

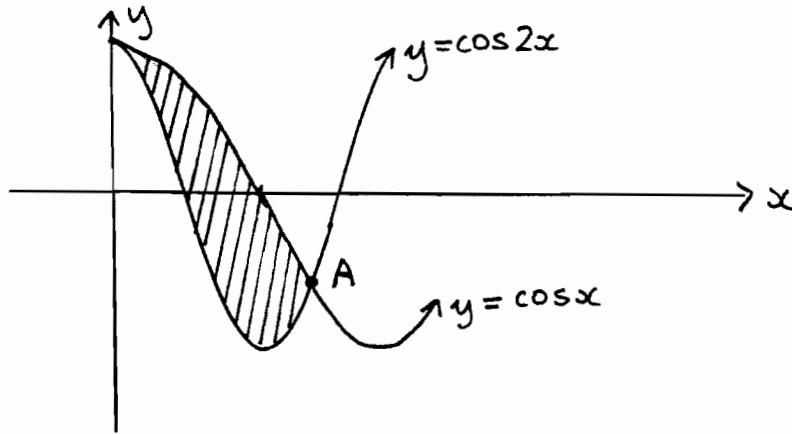


Show the area of the portion BDC is $2(4 - \pi)\text{cm}^2$

3

Question 4 (start a new page) (12 marks)

- a) The diagram shows parts of the curves $y = \cos x$ and $y = \cos 2x$



4

The coordinates of A are $(\frac{2\pi}{3}, -\frac{1}{2})$

Show that the shaded area is $\frac{3\sqrt{3}}{4} \text{ unit}^2$.

- b) Prove that the curve $y = x + 2 \cos x$ has a maximum turning point at $x = \frac{\pi}{6}$ in the domain $0 \leq x \leq \frac{\pi}{2}$ (do not sketch the curve).

4

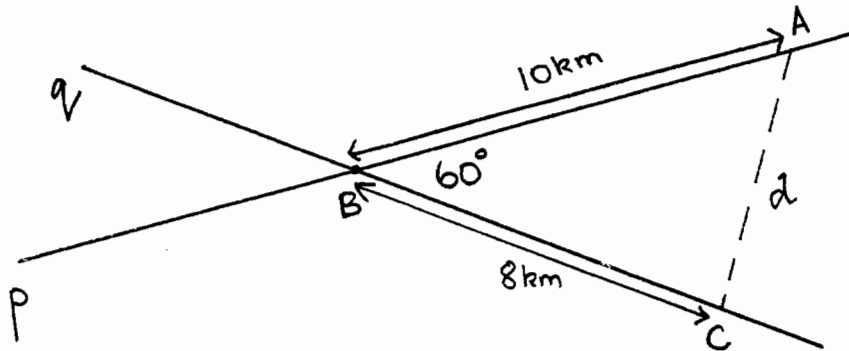
- c) i) Sketch the parabola $y = x^2 - x$ indicating where it cuts the x axis.
ii) The area enclosed by the parabola $y = x^2 - x$ and the x axis is rotated around the x axis. Find the volume of the solid generated.

1

3

Question 5 (start a new page) 10 marks

- a) A couple borrow \$320,000 at 6% p.a. The interest on the loan is compounded monthly on the balance owing. The loan is to be repaid in equal monthly instalments over 30 years. Let the monthly instalment be \$ M and the amount owing after n months be \$ A_n .
- Find an expression for \$ A_1 , the amount owing after one month. 1
 - Find the monthly instalment if the loan is to be fully repaid in 30 years. 4
- b) Two streets p and q intersect at B at an angle of 60° . Andrew is at A , 10 km from B and walks towards B at 5 km/h. Con is at C , 8 km from B and walks towards B at 6 km/h.



After t hours Andrew has walked $5t$ km towards B and Con has walked $6t$ km towards B .

- Use the cosine rule to show the distance d between Andrew and Con can be given by $d^2 = 31t^2 - 96t + 84$ 2
- Hence find how many hours (correct to 2 decimal places) until Andrew and Con are the least distance apart. 3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

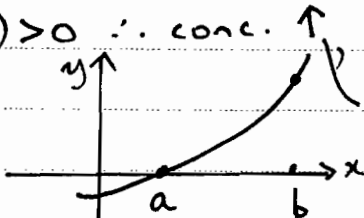
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

- a) $\sin \frac{2\pi}{3} = \sin(\pi - \frac{\pi}{3})$
 $= \sin \pi/3$
 $= \sqrt{3}/2$ (1)
- b) $\cos 1.5 = 0.071$ (1)
- c) $2.25\pi = 405^\circ$ (1) d) $1/2$ (1)
- e) $A = \int_a^c f(x) dx - \int_b^c g(x) dx$
 a (1) b (1)
- f) $a=3$ $\frac{2\pi}{b} = \pi \therefore b=2$
 $\therefore y = 3 \cos 2x$ (2)
- g) $f'(x) > 0 \therefore +ve$ gradient (1)
 $f''(x) > 0 \therefore$ conc. $\uparrow$ pt(a,0) (1)


Question 2

- a) $A_y = \int_0^2 (y^2 + 2) dy$
 $y = \sqrt{x-2}$
 $\therefore y^2 = x-2$
 $y^2 + 2 = x$
 $\therefore A_y = \left[\frac{y^3}{3} + 2y \right]_0^2$ (1)
 $= \frac{8}{3} + 4$ (1)
 $= 6\frac{2}{3} \text{ unit}^2$
- bi) $A = \frac{2}{2} [2 + 6 + 2(2)]$ (1)
 $= 12 \text{ unit}^2$ (1)
- ii) under estimate as area
 { from $x=1$ to $x=3$ becomes approximated to 4 and area above line $y=2$ is

- c) $V = \pi \int_0^{1/2} (\sqrt{\cos \pi x})^2 dx$ (1)
 $= \pi \int_0^{1/2} \cos \pi x dx$
 $= \pi \left[\frac{1}{\pi} \sin \pi x \right]_0^{1/2}$
 $= \sin \frac{\pi}{2} - \sin 0$ (1)
 $= 1 \text{ unit}^3$ (1)
- d) $\int_{\pi/6}^{\pi/3} \sec^2 x dx$ (1)
 $= [\tan x]_{\pi/6}^{\pi/3}$
 $= \tan \frac{\pi}{3} - \tan \frac{\pi}{6}$
 $= \sqrt{3} - \frac{1}{\sqrt{3}}$ (1)

Question 3

- a) $2\cos^2 x + 3\cos x - 2 = 0$
 $(2\cos x - 1)(\cos x + 2) = 0$
 $\cos x = \frac{1}{2}$ $\cos x = -2$
 $x = \frac{\pi}{3}, \frac{5\pi}{3}$ (1) / no solution
- b) i) $\frac{d}{dx} (\sin x)^2 = 2 \cos x \cdot \sin x$ (2)
- ii) $u = \sin x$ $v = \cos 2x$
 $u' = \cos x$ $v' = -2 \sin 2x$
 $\frac{dy}{dx} = \cos x \cdot \cos 2x - 2 \sin x \cdot \sin 2x$ (2)
- iii) $\int \sin(2x+1) dx$ (2)
 $= \frac{1}{2} \cos(2x+1) + C$
 do not take off mk if no "C"
- c) $\hat{A}BC = \frac{\pi}{4}$
 Area BDC = Area $\triangle ABC$ - sector ABD (1)

Question 4

a) $A = 2\pi/3$

$$\textcircled{1} \int_0^{2\pi/3} (\cos x - \cos 2x) dx$$

$$= \left[\sin x - \frac{1}{2} \sin 2x \right]_0^{2\pi/3}$$

$$\textcircled{1} = \sin \frac{2\pi}{3} - \frac{1}{2} \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2}$$

$$\textcircled{1} = \frac{2\sqrt{3} + \sqrt{3}}{4}$$

$$\textcircled{1} = \frac{3\sqrt{3}}{4} \text{ units}^2$$

b) $y = x + 2 \cos x$

$$\frac{dy}{dx} = 1 - 2 \sin x \quad \textcircled{1}$$

$$\frac{d^2y}{dx^2} = -2 \cos x$$

st pt $\frac{dy}{dx} = 0 \quad 1 - 2 \sin x = 0$

$$\therefore \sin x = 1/2$$

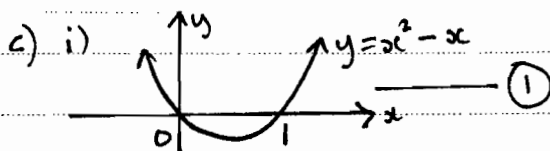
$$\textcircled{1} \quad x = \pi/6 \quad \text{if } 0 \leq x \leq \pi/2$$

test max/min for $x = \pi/6$

$$\frac{d^2y}{dx^2} = -2 \cos \frac{\pi}{6} < 0$$

$$\therefore \text{max} \quad \textcircled{1}$$

$$\therefore \text{max turning pt at } x = \frac{\pi}{6}$$



ii) $V = \pi \int_0^1 (x^2 - x)^2 dx$ $\textcircled{1}$

$$= \pi \int_0^1 (x^4 - 2x^3 + x^2) dx$$

$$= \pi \left[\frac{x^5}{5} - \frac{2x^4}{4} + \frac{x^3}{3} \right]_0^1 \quad \textcircled{1}$$

Question 5

a) \$320,000

6% pa $\Rightarrow$.5% p.m

30 yrs $\Rightarrow$ 360 months

i) $\$A_1 = 320,000 (1 + \frac{.5}{100})^1 - M$

$$= 320,000 (1.005)^1 - M \quad \textcircled{1}$$

ii) $\$A_2 = (320,000 (1.005) - M) 1.005 - M$

$$\textcircled{1} = 320,000 (1.005)^2 - 1.005M - M$$

$$\therefore$$

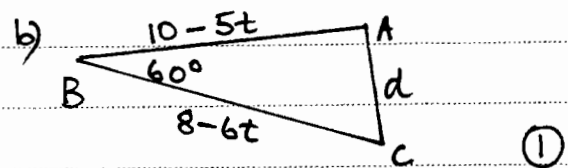
$$\$A_{360} = 320,000 (1.005)^{360} - M(1.005 + 1.005^2 + \dots + 1.005^{359})$$

$\textcircled{1}$ a.p. $a=1 \quad r=1.005 \quad n=360$

$\$A_{360} = 0$ as loan repaid

$$M \left[\frac{1.005^{360} - 1}{1.005 - 1} \right] = 320,000 (1.005^{360})$$

$$M = \$1918.56 \quad \textcircled{1}$$



i) $d^2 = (10 - 5t)^2 + (8 - 6t)^2 - 2(10 - 5t)(8 - 6t) \cos 60^\circ$

$$= 100 - 100t + 25t^2 + 64 - 96t + 36t^2 - (80 - 100t + 30t^2)$$

$$\therefore d^2 = 84 - 96t + 31t^2 \quad \textcircled{1}$$

ii) $\frac{d(d^2)}{dt} = -96 + 62t \quad \textcircled{1}$

$$\frac{d^2(d^2)}{dt^2} = 62 > 0 \quad \therefore \text{minimum}$$

st pt $-96 + 62t = 0 \quad \therefore t = 1.548 \text{ hrs}$

$$\therefore t = 1.55 \text{ hr (2 dec pl)} \quad \textcircled{1}$$

Question 1

a) $\sin \frac{2\pi}{3} = \sin(\pi - \frac{\pi}{3})$
 $= \sin \frac{\pi}{3}$
 $= \frac{\sqrt{3}}{2}$

b) $\cos 1.5 = 0.071$

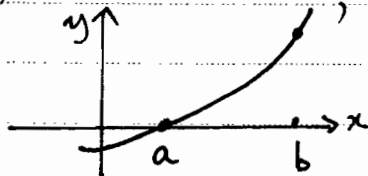
c) $2.25\pi = 405^\circ$ d) $\frac{1}{2}$

e) $A = \int_a^c f(x) dx - \int_b^c g(x) dx$

f) $a=3$ $\frac{2\pi}{b} = \pi \therefore b=2$

$\therefore y = 3 \cos 2x$

g) $f'(x) > 0 \therefore +ve$ gradient
 $f''(x) > 0 \therefore$ conc. $\uparrow$, pt(a,0)



Question 2

a) $A_y = \int_0^2 (y^2 + 2) dy$

$y = \sqrt{x-2}$

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$y^2 + 2 = x$

$\therefore A_y = \left[\frac{y^3}{3} + 2y \right]_0^2$

$= \frac{8}{3} + 4$

$= 6\frac{2}{3} \text{ unit}^2$

bi) $A = \frac{2}{2} [2 + 6 + 2(2)]$
 $= 12 \text{ unit}^2$

ii) under estimate as area

from $x=1$ to $x=3$ becomes approximated to 4 and area above line $y=2$ is

c) $V = \pi \int_0^{1/2} (\sqrt{\cos \pi x})^2 dx$
 $= \pi \int_0^{1/2} \cos \pi x dx$
 $= \pi \left[\frac{1}{\pi} \sin \pi x \right]_0^{1/2}$
 $= \sin \frac{\pi}{2} - \sin 0$
 $= 1 \text{ unit}^3$

d) $\int_{\pi/6}^{\pi/3} \sec^2 x dx$
 $= \left[\tan x \right]_{\pi/6}^{\pi/3}$
 $= \tan \frac{\pi}{3} - \tan \frac{\pi}{6}$
 $= \sqrt{3} - \frac{1}{\sqrt{3}}$

Question 3

a) $2\cos^2 x + 3\cos x - 2 = 0$
 $(2\cos x - 1)(\cos x + 2) = 0$
 $\cos x = \frac{1}{2} \quad \cos x = -2$
 $x = \frac{\pi}{3}, \frac{5\pi}{3} \quad \text{no solution}$

b) i) $\frac{d}{dx} (\sin x)^2 = 2 \cos x \cdot \sin x$

ii) $u = \sin x \quad v = \cos 2x$
 $u' = \cos x \quad v' = -2 \sin 2x$

$\frac{dy}{dx} = \cos x \cdot \cos 2x - 2 \sin x \cdot \sin 2x$

iii) $\int \sin(2x+1) dx$
 $= \frac{1}{2} \cos(2x+1) + C$

c) $\hat{ABC} = \frac{\pi}{4}$

Area BDC = Area $\triangle ABC$ - sector ABD

Question 4

$$\begin{aligned}
 \text{a) } A &= 2\pi/3 \int_0^{2\pi/3} (\cos x - \cos 2x) dx \\
 &= \left[\sin x - \frac{1}{2} \sin 2x \right]_0^{2\pi/3} \\
 &= \sin \frac{2\pi}{3} - \frac{1}{2} \sin \frac{\pi}{3} \\
 &= \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \\
 &= \frac{2\sqrt{3} + \sqrt{3}}{4} \\
 &= \frac{3\sqrt{3}}{4} \text{ units}^2
 \end{aligned}$$

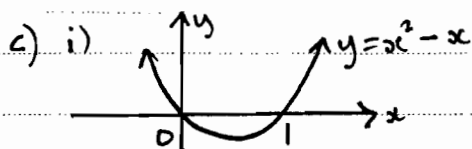
$$\begin{aligned}
 \text{b) } y &= x + 2 \cos x \\
 \frac{dy}{dx} &= 1 - 2 \sin x \\
 \frac{d^2y}{dx^2} &= -2 \cos x
 \end{aligned}$$

$$\begin{aligned}
 \text{st pt } \frac{dy}{dx} &= 0 \quad 1 - 2 \sin x = 0 \\
 \therefore \sin x &= 1/2 \\
 x &= \pi/6 \quad \text{if } 0 \leq x \leq \pi/2
 \end{aligned}$$

test max/min for $x = \pi/6$

$$\begin{aligned}
 \frac{d^2y}{dx^2} &= -2 \cos \frac{\pi}{6} < 0 \\
 \therefore \text{max}
 \end{aligned}$$

$\therefore$ max turning pt at $x = \frac{\pi}{6}$



$$\begin{aligned}
 \text{ii) } V &= \pi \int_0^1 (x^2 - x)^2 dx \\
 &= \pi \int_0^1 (x^4 - 2x^3 + x^2) dx \\
 &= \pi \left[\frac{x^5}{5} - \frac{2x^4}{4} + \frac{x^3}{3} \right]_0^1
 \end{aligned}$$

Question 5

a) \$320,000

6% pa $\Rightarrow$.5% p.m

30 yrs $\Rightarrow$ 360 months

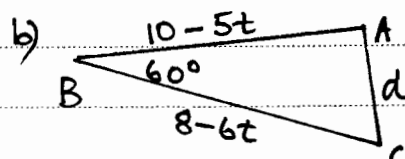
$$\begin{aligned}
 \text{i) } \$A_1 &= 320,000 (1 + \frac{.5}{100})^1 - M \\
 &= 320,000 (1.005)^1 - M
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \$A_2 &= (320,000 (1.005) - M) 1.005 - M \\
 &= 320,000 (1.005)^2 - 1.005M - M
 \end{aligned}$$

$$\begin{aligned}
 \therefore \\
 \$A_{360} &= 320,000 (1.005)^{360} - M(1 + 1.005 + \dots + 1.005^{359})
 \end{aligned}$$

G.P. $a=1$ $r=1.005$ $n=360$
 $\$A_{360} = 0$ as loan repaid

$$\begin{aligned}
 M \left[\frac{1(1.005^{360} - 1)}{1.005 - 1} \right] &= 320,000 (1.005)^{360} \\
 M &= \$1918.56
 \end{aligned}$$



$$\begin{aligned}
 \text{i) } d^2 &= (10-5t)^2 + (8-6t)^2 - 2(10-5t)(8-6t) \cos 60^\circ \\
 &= 100 - 100t + 25t^2 + 64 - 96t + 36t^2 \\
 &\quad - (80 - 100t + 36t^2)
 \end{aligned}$$

$$\therefore d^2 = 84 - 96t + 36t^2$$

$$\text{ii) } \frac{d(d^2)}{dt} = -96 + 72t$$

$$\frac{d^2(d^2)}{dt^2} = 72 > 0 \quad \therefore \text{minimum}$$

$$\begin{aligned}
 \text{st pt } -96 + 72t &= 0 \quad \therefore t = 1.548 \text{ hrs} \\
 \therefore t &= 1.55 \text{ hr (2 dec pl)}
 \end{aligned}$$